

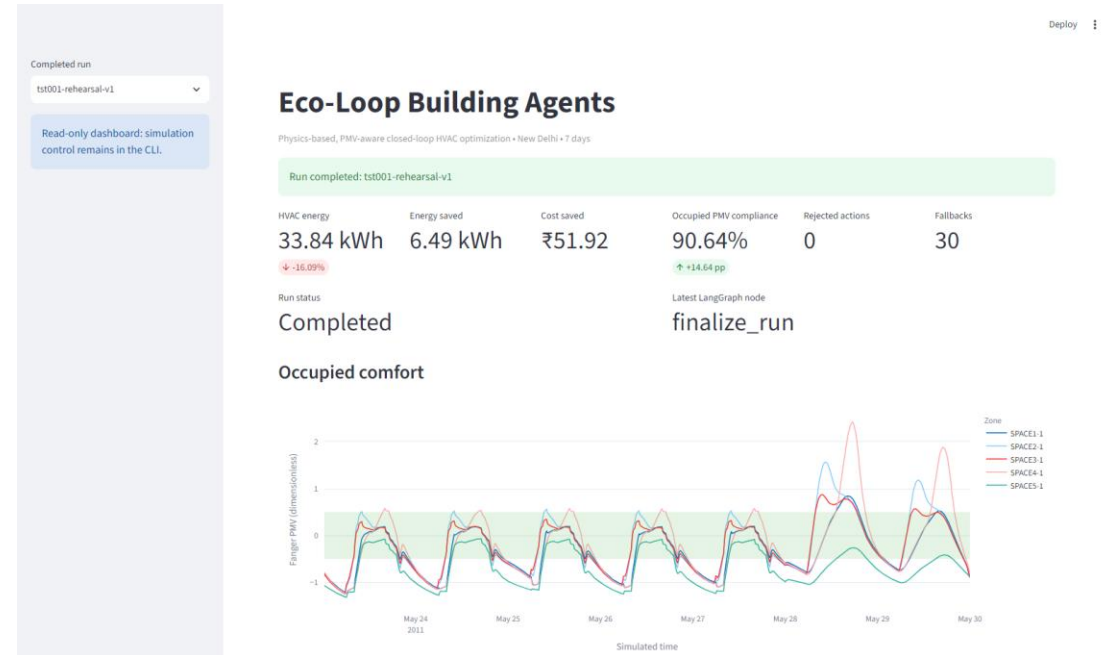
ECO-LOOP BUILDING AGENTS

- **PS ID – Question 1**
- **PS Title – Eco-Loop Building Agents**
- **Theme – Sustainable Buildings**
- **Category – Software**
- **Student – Gokulan Malli Jayaprakash**
- **ID – C10421040**

ECO-LOOP BUILDING AGENTS

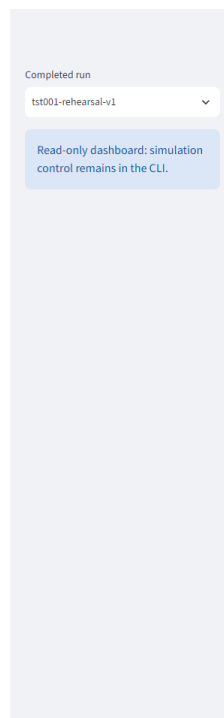
Proposed Solution

- EnergyPlus streams five-zone temperature, occupancy, PMV and HVAC energy.
- LangGraph runs Energy → Comfort → Supervisor → Safety → Action → Reflection.
- Measured: 16.09% less HVAC energy; occupied comfort rises from 76.00% to 90.64%.



TECHNICAL APPROACH

- Stack: Python 3.12, EnergyPlus 26.1, LangGraph, FastMCP, Pydantic, Streamlit and Qwen3 4B.
- Loop: observe → agents → safety → server reauthorization → actuation → measured reflection.



Baseline vs controlled cumulative HVAC electricity



Latest agent cycle

Safe structured outputs reconstructed from the persisted audit.

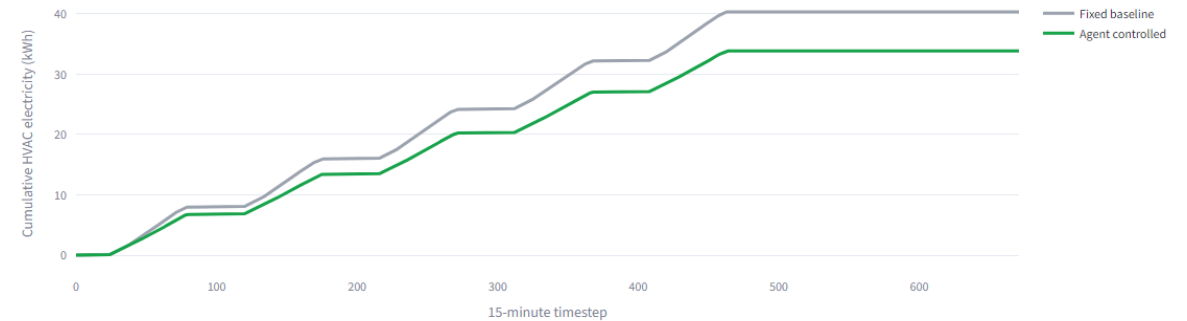
role	structured_output
Energy	setpoint=25.50°C; expected_energy_effect=NEU
Comfort	occupancy=0; risk=LOW
Supervisor	disposition=ACCEPT; authorized_setpoint=25.5
Reflection	outcome=SAFE; recommend_continue=false

Decision and reflection chronology

observation_sequence	decision_id	phase	Requested setpoint (°C)	Applied setpoint
168	tst001-rehearsal-v1-decision-0168	proposed → applied	25.5	
167	tst001-rehearsal-v1-decision-0167	proposed → applied	25.5	
166	tst001-rehearsal-v1-decision-0166	proposed → applied	25.5	
165	tst001-rehearsal-v1-decision-0165	proposed → applied	25.5	
164	tst001-rehearsal-v1-decision-0164	proposed → applied	25.5	
163	tst001-rehearsal-v1-decision-0163	proposed → applied	25.5	
162	tst001-rehearsal-v1-decision-0162	proposed → applied	25.5	
161	tst001-rehearsal-v1-decision-0161	proposed → applied	25.5	
160	tst001-rehearsal-v1-decision-0160	proposed → applied	25.5	
159	tst001-rehearsal-v1-decision-0159	proposed → applied	25.5	

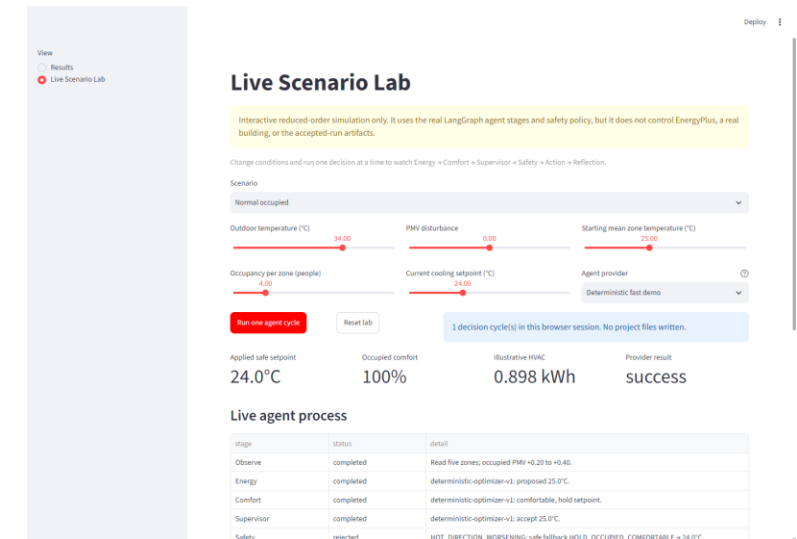
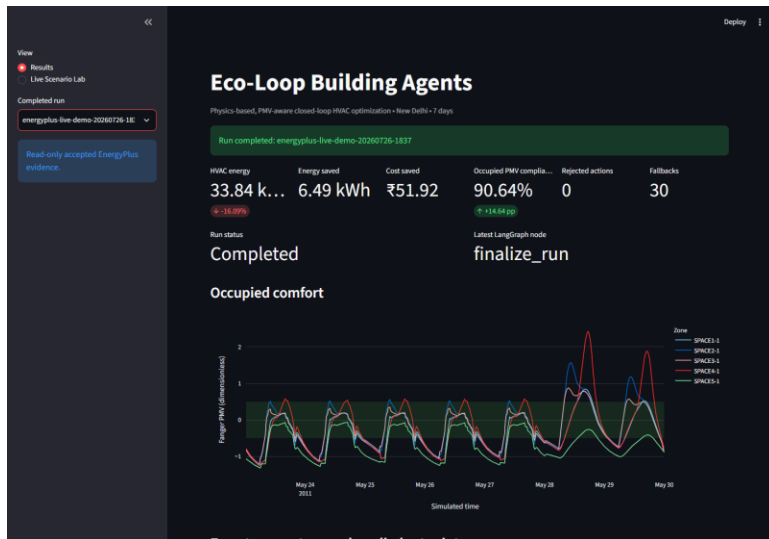
FEASIBILITY AND VIABILITY

- Proven locally: 672 timesteps, 168/168 actions, 6.05 s and zero severe errors.
- Risks: unsafe output, stale actions, timeout, comfort conflict and irreproducible runs.
- Mitigation: PMV guard, 22–28°C bounds, 1°C limit, double authorization and immutable runs.



ARTIFACTS

- Working proof:
 - Source, IDF, lockfile and 377 tests
 - Real EnergyPlus, PMV and agent evidence
 - Isolated Live Scenario Lab



RESEARCH AND REFERENCES

- EnergyPlus Python API — energyplus.readthedocs.io/en/latest/api.html
- Fanger PMV — bigladdersoftware.com/epx/docs/25-1/input-output-reference/
- LangGraph — github.com/langchain-ai/langgraph
- Model Context Protocol Python SDK — github.com/modelcontextprotocol/python-sdk
- Ollama structured chat — docs.ollama.com/api/chat
- Project evidence — controlled-run001-optimized-v3; 377 tests; 90.47% branch coverage
- Honest boundary — accepted metrics use the deterministic optimizer; Qwen3 4B is advisory and cannot bypass safety.